

Engineering Frontier AI Workshop

Workshop Flow

- **Session 1:** Evolution of Modern AI
- **Session 2:** Machine Learning Fundamentals
- **Session 3:** Introduction to Large Language Models
- **Session 4:** Reinforcement Learning from Verifiable Rewards
- **Lab 1:** Introduction to AI Training Environment
- **Lab 2:** Cold start SFT dataset -> Cold start SFT -> GRPO dataset -> GRPO Training
-> Evaluations -> Distillation Training -> Evaluations.
- **Session 5:** Introduction to Vision Language Models
- **Lab 3:** Projector Pre Training Dataset -> Projector Pretraining -> SFT dataset -> SFT
-> Inference.
- **Session 6:** The Future of AI

Model	Params	MMLU / Redux	HellaSwag (acc/ acc_norm)	ARC-Challenge (acc/acc_norm)	GSM8K	GPQA	IFEval (Strict)
Lily 1.5B v0.3	1.5B	55.16 / 61.05	46.92 / 59.41	34.22 / 37.37	62.70	13.62	44.36
Qwen2.5-1.5B-Instruct	1.5B	— / 50.7	—	—	73.2	29.8	42.5
Google Gemma 3 1B	1B	59.6	62.3	38.4	38.4	15.0	—
Meta Llama 3.2 1B Instruct	1B	~49	~60	~35	~44	~11	~34
Hugging Face SmolLM2-1.7B-Instruct	1.7B	~53	~69	~46	~59	~14	~38